

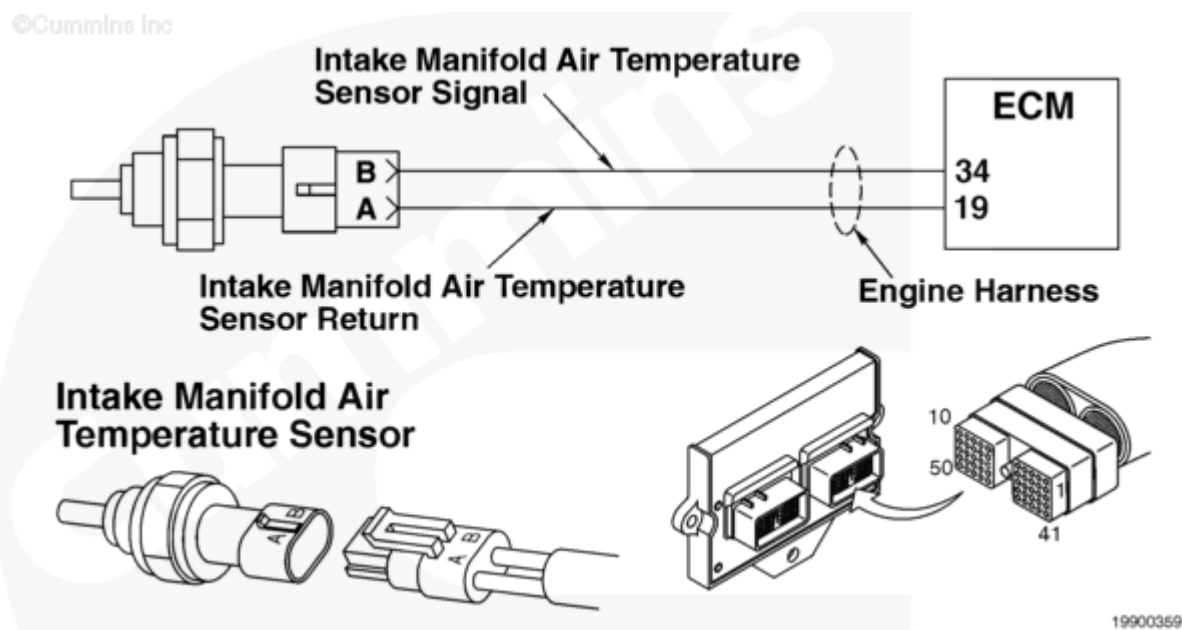
Fault Code: 153

Intake Manifold Temperature Sensor Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 153 PID(P): P105 SPN: 105 FMI: 3 Lamp: Yellow SRT:	More than 4.95 VDC detected at the intake manifold temperature sensor signal pin 34 of the engine harness electronic control module (ECM) connector.	Engine protection for intake manifold air temperature is disabled.



Intake Manifold Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is used by the ECM to monitor the temperature of the air in the intake manifold after the aftercooler. The intake manifold temperature sensor is used by the ECM for the engine protection system, timing, and fueling control.

The ECM monitors the voltage on pin 34. The ECM expects to see the voltage vary between 0.5 and 4.5 VDC. If the voltage is above 4.95 VDC, the ECM will log Fault Code 153.

Voltage above 4.95 VDC on pin 34 can be caused by opens in the signal or return wires, voltage short circuits to the signal or return wires, or a failed open sensor.

Component Location

Two intake manifold temperature sensors are used on the QST30 industrial engine, one on each side. The sensors are located in the intake manifold at the rear of the engine.

Shoptalk

The resistance of all the temperature sensors varies with the temperature. The reading you observe should compare to the following table if the sensor is functioning properly.

Temperature (°C)	Temperature [°F]	Resistance (ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-153 (</qs3/pubsys2/xml/en/procedures/87/87-t05-153.html>)